Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) building backend services and large-scale data pipelines with Python, AWS, and Kubernetes. Experience designing ETL/data-processing frameworks, containerized microservices, and event-driven systems that ingest, process, and serve large datasets reliably.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, Go, JavaScript, TypeScript, C/C++, SQL.

Data & Pipelines: PySpark, Delta Lake, Apache Spark, ETL Design, pandas, NumPy, PostgreSQL, MySQL, Redis.

Backend & APIs: FastAPI, Flask, Django, Node.js, REST APIs, Microservices, Event-Driven Systems.

Cloud & Infra: AWS (S3, EC2, Lambda, SQS), Docker, Kubernetes, ArgoCD, Helm, Terraform, GitLab CI/CD, Jenkins, Git, Linux, Bash.

AI & ML: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain.

WORK EXPERIENCE

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**; developed **Go + Terratest** framework catching **95%** of misconfigurations pre-production.

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps, demonstrating **25.7x latency reduction**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built end-to-end data platform ingesting and processing **2.3M-row** taxi/rideshare dataset via Spark-based medallion (bronze/silver/gold) pipeline across **263 zones**; served real-time predictions through a FastAPI backend with PostgreSQL and Redis caching.
- Designed two-stage Poisson hurdle demand model, improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

Spotify Personal Analytics Platform [Python, FastAPI, PostgreSQL, ETL] Jul 2026

- Engineered 4-stage Python ETL pipeline processing **71K+** streaming events with custom rate limiting and exponential-backoff retry handling; built checkpointed ingestion stage processing **13.8K** tracks with multi-source API fallback.
- Designed PostgreSQL analytics schema with materialized views, composite/partial indexes, and **16 SQL RPC functions** for low-latency pre-aggregated queries over streaming history.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).